

Wazuh Deployment & Custom Rule Engineering

1. Objective

The objective of this activity is to deploy a central Wazuh Manager on an Ubuntu VM and connect a Wazuh Agent (Kali Linux) to it. After establishing the connection, we will engineer Custom Detection Rules (Level 10) to monitor high-risk system commands like whoami and apt install.

2. Lab Environment

- **Wazuh Manager:** Ubuntu 22.04 VM (Central Monitoring Server).
- **Wazuh Agent:** Kali Linux VM (Target machine being monitored).
- **Connectivity:** Both machines must be on the same network (NAT/Bridged) to communicate via IP.

3. Step-by-Step Procedure

Step 1: Wazuh Manager Installation (Machine: Ubuntu)

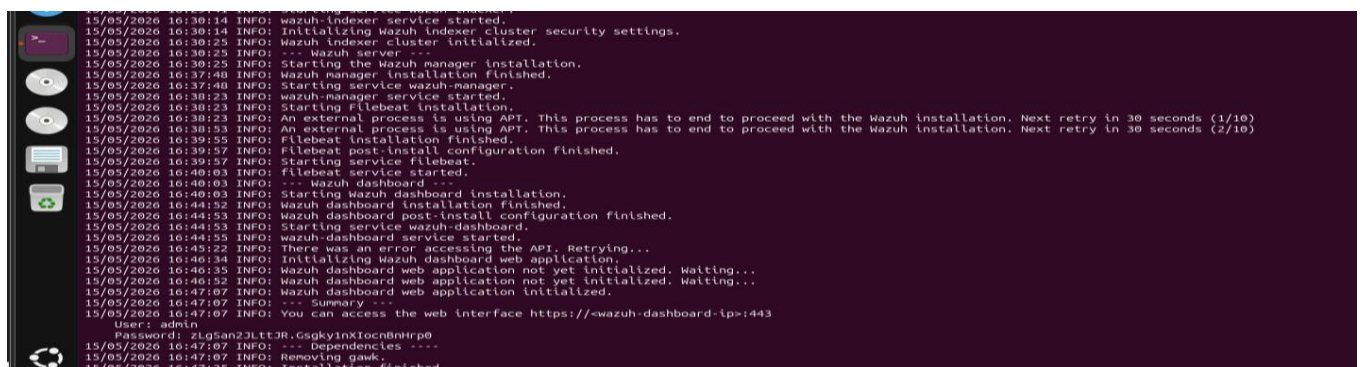
The Wazuh Manager is the central brain of the SIEM, responsible for collecting logs and processing security rules. For this lab, the manager was deployed on an Ubuntu VM using a forced installation method to ensure full stack compatibility.

1. **Forced Installation:** The installation was triggered using the `-i` flag to bypass system compatibility checks and deploy the Indexer, Server, and Dashboard components.

Command: `sudo bash wazuh-install.sh -a -i`

2. **Credential Retrieval:** After the process was completed, the terminal provided the unique **Admin Password**. This was saved immediately to allow access to the web interface.
3. **IP Identification:** The Manager's network identity was retrieved using the `hostname -I` command to prepare for the agent connection phase.

Manager IP: 192.168.5.131



```
15/05/2026 16:30:14 INFO: wazuh-indexer service started.
15/05/2026 16:30:14 INFO: Initializing Wazuh indexer cluster security settings.
15/05/2026 16:30:23 INFO: Wazuh indexer cluster initialized.
15/05/2026 16:30:25 INFO: --- Wazuh server ---
15/05/2026 16:30:25 INFO: Starting the Wazuh manager installation.
15/05/2026 16:37:40 INFO: Wazuh manager installation finished.
15/05/2026 16:37:40 INFO: Starting service wazuh-manager.
15/05/2026 16:38:23 INFO: wazuh-manager service started.
15/05/2026 16:38:23 INFO: Starting filebeat installation.
15/05/2026 16:38:23 INFO: An external process is using API. This process has to end to proceed with the Wazuh installation. Next retry in 30 seconds (1/10)
15/05/2026 16:38:53 INFO: An external process is using API. This process has to end to proceed with the Wazuh installation. Next retry in 30 seconds (2/10)
15/05/2026 16:39:05 INFO: Filebeat installation finished.
15/05/2026 16:39:05 INFO: Starting service filebeat.
15/05/2026 16:39:05 INFO: Filebeat service started.
15/05/2026 16:40:03 INFO: --- Wazuh dashboard ---
15/05/2026 16:40:03 INFO: Starting Wazuh dashboard installation.
15/05/2026 16:44:53 INFO: Wazuh dashboard post-install configuration finished.
15/05/2026 16:44:53 INFO: Starting service wazuh-dashboard.
15/05/2026 16:44:53 INFO: Wazuh dashboard service started.
15/05/2026 16:45:22 INFO: There was an error accessing the API. Retrying...
15/05/2026 16:46:34 INFO: Initializing Wazuh dashboard web application.
15/05/2026 16:46:35 INFO: Wazuh dashboard web application not yet initialized. Waiting...
15/05/2026 16:46:52 INFO: Wazuh dashboard web application not yet initialized. Waiting...
15/05/2026 16:47:07 INFO: Wazuh dashboard web application initialized.
15/05/2026 16:47:07 INFO: --- Summary ---
15/05/2026 16:47:07 INFO: You can access the web interface https://wazuh-dashboard-ip:443
User: admin
Password: 2Lg5an23Ltt3R.6gkyinKioCnBnirp0
15/05/2026 16:47:07 INFO: --- Dependencies ---
15/05/2026 16:47:07 INFO: Removing gawk.
15/05/2026 16:47:25 INFO: Installation finished.
```

```
moiz@moiz-VMware-Virtual-Platform:~$ hostname -I
192.168.5.131
moiz@moiz-VMware-Virtual-Platform:~$
```

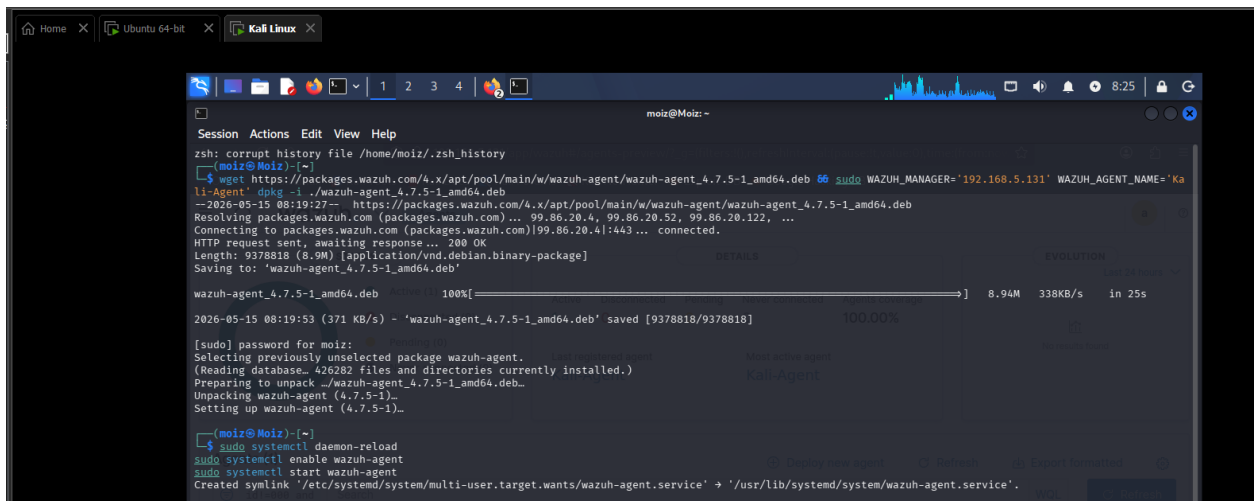
Observation: The Manager's IP address (192.168.5.131) was identified, which is a critical requirement for linking the Kali Linux agent in the next phase.

Step 2: Wazuh Agent Installation & Connection (Machine: Kali Linux)

In this phase, the "Worker" (Agent) is deployed on the Kali Linux machine and linked to the central "Boss" (Manager). This connection is vital for the Manager to receive telemetry and security events from the endpoint.

1. **Agent Deployment Configuration:** Access the Wazuh Dashboard via browser at <https://192.168.5.131>. Navigate to **Agents > Deploy new agent**, select **Linux (DEB amd64)**, and enter the Manager's IP address.
2. **Downloading and Installing the Agent:** The generated command was executed in the Kali terminal to download the .deb package and install the Wazuh agent.

Command: `wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb && sudo WAZUH_MANAGER='192.168.5.131' WAZUH_AGENT_NAME='Kali-Agent' dpkg -i ./wazuh-agent_4.7.5-1_amd64.deb`



```
moiz@moiz: ~
Session Actions Edit View Help
zsh: corrupt history file /home/moiz/.zsh_history
moiz@moiz:~$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb && sudo WAZUH_MANAGER='192.168.5.131' WAZUH_AGENT_NAME='Kali-Agent' dpkg -i ./wazuh-agent_4.7.5-1_amd64.deb
--2026-05-15 08:19:27-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 99.86.20.4, 99.86.20.52, 99.86.20.122, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|99.86.20.4|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9378818 (8.9M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.7.5-1_amd64.deb'

wazuh-agent_4.7.5-1_amd64.deb 100%[=====] 8.94M 338KB/s in 25s

2026-05-15 08:19:53 (371 KB/s) - 'wazuh-agent_4.7.5-1_amd64.deb' saved [9378818/9378818]

[sudo] password for moiz:
Selecting previously unselected package wazuh-agent.
(Reading database... 426282 files and directories currently installed.)
Preparing to unpack ./wazuh-agent_4.7.5-1_amd64.deb...
Unpacking wazuh-agent (4.7.5-1)...
Setting up wazuh-agent (4.7.5-1)...

moiz@moiz:~$ sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' → '/usr/lib/systemd/system/wazuh-agent.service'.
```

Observation: The agent package was successfully downloaded from the official Wazuh repository and installed locally on the Kali machine with the manager IP pre-configured.

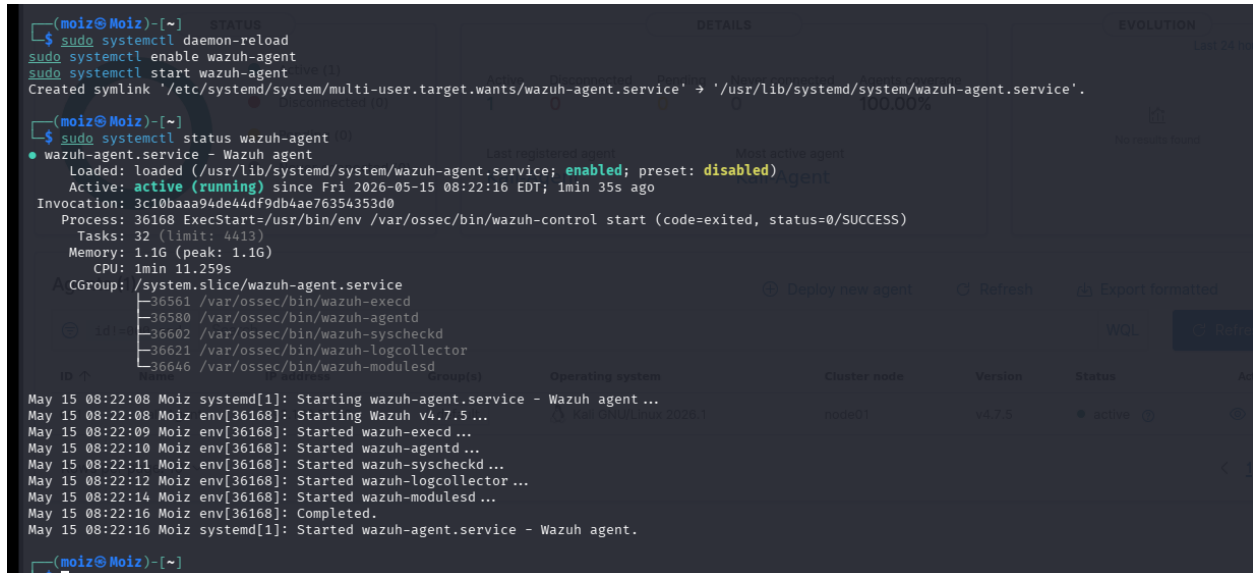
3. **Service Initialization:** After installation, the system daemon was reloaded, and the agent service was enabled and started to establish the handshake.

Commands: `sudo systemctl daemon-reload`
`sudo systemctl enable wazuh-agent`

sudo systemctl start wazuh-agent

4. **Verification of Connection:** The final status was checked on the Kali machine to ensure the service was operational.

Command: sudo systemctl status wazuh-agent



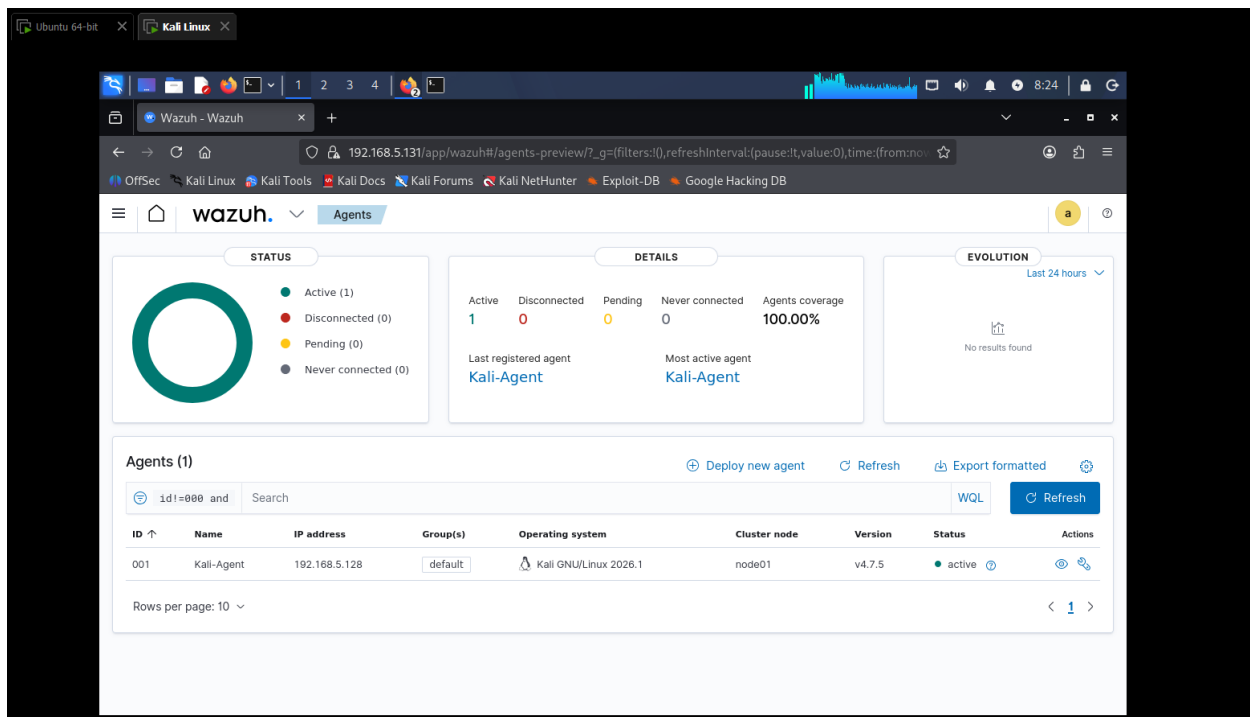
```
(moiz@Moiz)-[~]
$ sudo systemctl daemon-reload
$ sudo systemctl enable wazuh-agent
$ sudo systemctl start wazuh-agent
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' → '/usr/lib/systemd/system/wazuh-agent.service'.

(moiz@Moiz)-[~]
$ sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-05-15 08:22:16 EDT; 1min 35s ago
     Invocation: 3c10baaa94de44df9db4ae76354353d0
    Process: 36168 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
       Tasks: 32 (limit: 4413)
      Memory: 1.1G (peak: 1.1G)
         CPU: 1min 11.259s
    CGroup: /system.slice/wazuh-agent.service
            └─36561 /var/ossec/bin/wazuh-execd
              └─36580 /var/ossec/bin/wazuh-agentd
                └─36602 /var/ossec/bin/wazuh-syscheckd
                  └─36621 /var/ossec/bin/wazuh-logcollector
                    └─36646 /var/ossec/bin/wazuh-modulesd

May 15 08:22:08 Moiz systemd[1]: Starting wazuh-agent.service - Wazuh agent...
May 15 08:22:08 Moiz env[36168]: Starting Wazuh v4.7.5...
May 15 08:22:09 Moiz env[36168]: Started wazuh-execd...
May 15 08:22:10 Moiz env[36168]: Started wazuh-agentd...
May 15 08:22:11 Moiz env[36168]: Started wazuh-syscheckd...
May 15 08:22:12 Moiz env[36168]: Started wazuh-logcollector...
May 15 08:22:14 Moiz env[36168]: Started wazuh-modulesd...
May 15 08:22:16 Moiz env[36168]: Completed.
May 15 08:22:16 Moiz systemd[1]: Started wazuh-agent.service - Wazuh agent.
```

Observation: The active (running) status confirms that the agent is successfully communicating with the manager and the encryption keys have been exchanged.

5. **Manager-Side Dashboard Verification:** Finally, the Wazuh Dashboard was checked to confirm the endpoint is listed under active agents.



***Observation:** The central dashboard confirms that the Kali-Agent is officially enrolled and the manager is now ready to process incoming logs for custom rule analysis.*

Step 3: Custom Rule Engineering & Agent Log Forwarding

This stage involves configuring the communication between the Agent and the Manager. First, the Kali Agent must be instructed to forward its terminal logs; then, the Ubuntu Manager must be programmed with custom logic to identify specific forbidden commands.

Part A: Agent-Side Configuration (Machine: Kali Linux)

The Wazuh Agent must be explicitly instructed to monitor and forward execution logs to the central Manager for analysis.

1. **Modifying Configuration:** The agent configuration file was accessed to define the log source. **Command:** `sudo nano /var/ossec/etc/ossec.conf`
2. **Applying Logs Block:** The following XML block was added to monitor the Auditd log file:

XML

```
<localfile>
```

```
<log_format>audit</log_format>
```

```
<location>/var/log/audit/audit.log</location>
```

</localfile>

3. **Service Restart:** The agent was restarted to initialize the new logging parameters.

Command: sudo systemctl restart wazuh-agent

4. **Verification:** The service status was checked to confirm the agent is running without syntax errors. **Command:** sudo systemctl status wazuh-agent

```
(moiz@Moiz)-[~]
$ sudo truncate -s 0 /var/ossec/etc/ossec.conf
(moiz@Moiz)-[~]
$ sudo nano /var/ossec/etc/ossec.conf
(moiz@Moiz)-[~]
$ sudo systemctl restart wazuh-agent
(moiz@Moiz)-[~]
$ sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-05-15 09:44:34 EDT; 49s ago
     Invocation: 5e9a2d8c83a34122a0b27839205a9600
       Process: 80349 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
        Tasks: 23 (limit: 4413)
      Memory: 9.1M (peak: 10.8M)
         CPU: 61ms
    CGroup: /system.slice/wazuh-agent.service
           └─┬─80371 /var/ossec/bin/wazuh-execd
              ├──80390 /var/ossec/bin/wazuh-agentd
              ├──80412 /var/ossec/bin/wazuh-syscheckd
              ├──80423 /var/ossec/bin/wazuh-logcollector
              └─80448 /var/ossec/bin/wazuh-modulesd

May 15 09:44:29 Moiz env[80349]: Started wazuh-execd...
May 15 09:44:30 Moiz env[80349]: Started wazuh-agentd...
May 15 09:44:30 Moiz env[80410]: 2026/05/15 09:44:30 wazuh-syscheckd: INFO: (6678): No directory provided for syscheck to monitor. active
May 15 09:44:30 Moiz env[80410]: 2026/05/15 09:44:30 wazuh-syscheckd: INFO: (6001): File integrity monitoring disabled.
May 15 09:44:30 Moiz env[80410]: 2026/05/15 09:44:30 rootcheck: INFO: Rootcheck disabled.
May 15 09:44:30 Moiz env[80349]: Started wazuh-syscheckd...
May 15 09:44:31 Moiz env[80349]: Started wazuh-logcollector...
May 15 09:44:32 Moiz env[80349]: Started wazuh-modulesd...
May 15 09:44:34 Moiz env[80349]: Completed.
May 15 09:44:34 Moiz systemd[1]: Started wazuh-agent.service - Wazuh agent.
```

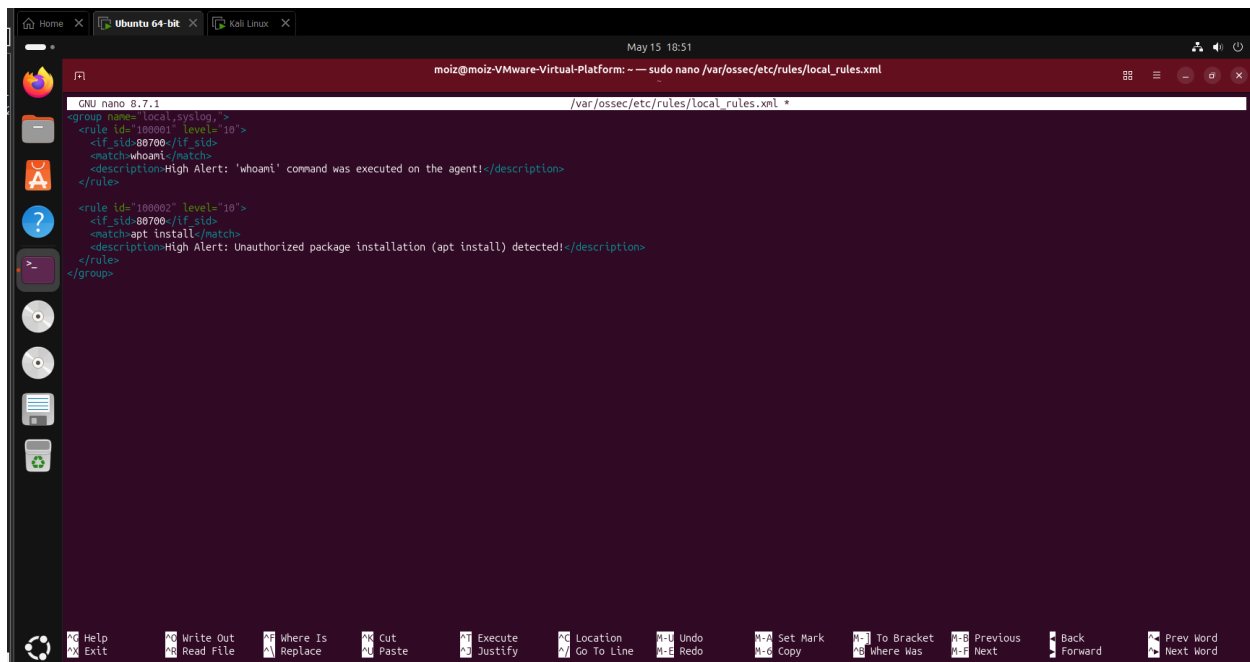
Observation: The "active (running)" status confirms that the log forwarding configuration is valid. The agent is now successfully streaming terminal execution data to the Manager.

Part B: Manager-Side Rule Engineering (Machine: Ubuntu Manager)

With the logs now arriving at the Manager, custom detection rules are required to trigger alerts for specific strings like whoami and apt install.

- **Step 1:** Open the local rules repository on the Manager.
 - **Command:** sudo nano /var/ossec/etc/rules/local_rules.xml

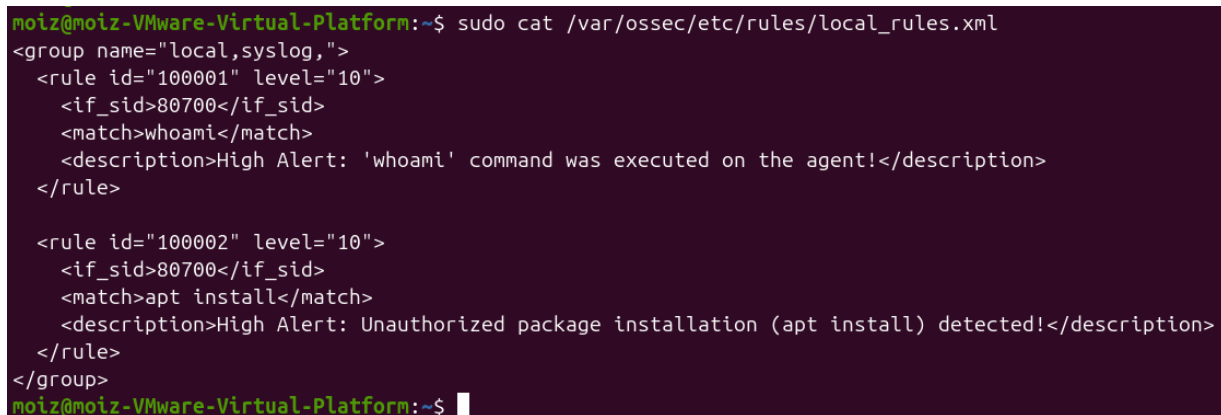
Step 2: Implement Rule IDs **100001** and **100002**. These are configured as children of the standard Auditd SID (80700) with a **Level 10** severity rating.



```
moiz@moiz-VMware-Virtual-Platform: ~ -- sudo nano /var/ossec/etc/rules/local_rules.xml
GNU nano 2.7.1 /var/ossec/etc/rules/local_rules.xml
<group name="local,syslog,">
  <rule id="100001" level="10">
    <if_sid>80700</if_sid>
    <match>whoami</match>
    <description>High Alert: 'whoami' command was executed on the agent!</description>
  </rule>
  <rule id="100002" level="10">
    <if_sid>80700</if_sid>
    <match>apt install</match>
    <description>High Alert: Unauthorized package installation (apt install) detected!</description>
  </rule>
</group>
```

Observation: The rules utilize the `<match>` tag to scan incoming logs for specific command strings. Assigning Level 10 severity ensures these events are categorized as high-priority security alerts on the dashboard.

- **Step 3:** Verify the file content and syntax integrity after saving.
 - **Command:** `sudo cat /var/ossec/etc/rules/local_rules.xml`



```
moiz@moiz-VMware-Virtual-Platform:~$ sudo cat /var/ossec/etc/rules/local_rules.xml
<group name="local,syslog,">
  <rule id="100001" level="10">
    <if_sid>80700</if_sid>
    <match>whoami</match>
    <description>High Alert: 'whoami' command was executed on the agent!</description>
  </rule>
  <rule id="100002" level="10">
    <if_sid>80700</if_sid>
    <match>apt install</match>
    <description>High Alert: Unauthorized package installation (apt install) detected!</description>
  </rule>
</group>
moiz@moiz-VMware-Virtual-Platform:~$
```

Observation: The `cat` command confirms that the XML structure is valid and the rules are correctly placed within the `<group>` tags, ready for the analysis engine.

- **Step 4:** Restart the Wazuh Manager to compile and activate the new ruleset.
 - **Command:** `sudo systemctl restart wazuh-manager`

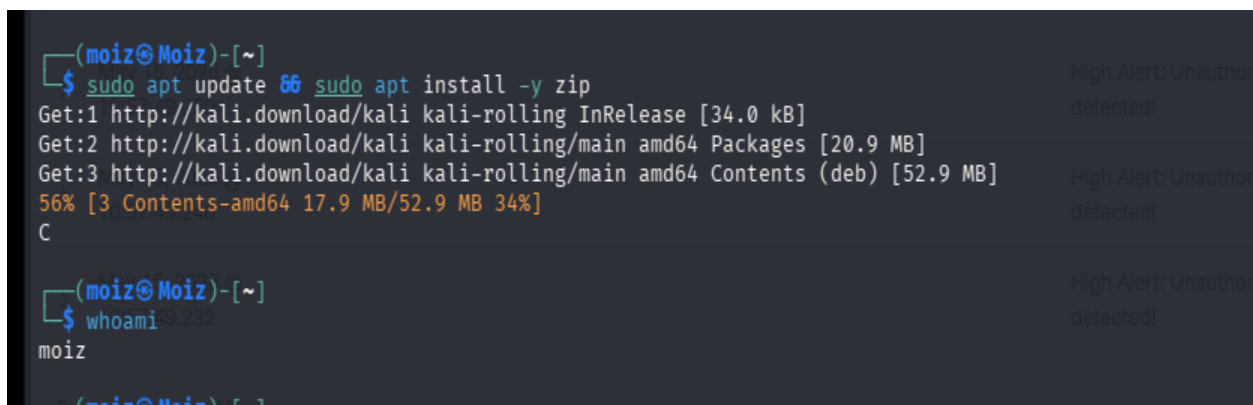
Step 4: Alert Validation & Final Monitoring

This final phase confirms the end-to-end success of the activity. By performing forbidden actions on the Kali Agent, we verify that the Manager correctly identifies these system interactions and alerts the security team via the dashboard.

1. Executing Forbidden Actions (Machine: Kali Agent)

The detection engine was tested by executing the reconnaissance and package management commands on the Kali Linux terminal.

- **Command 1:** whoami (To test Rule 100001).
- **Command 2:** sudo apt update && sudo apt install -y zip (To test Rule 100002).



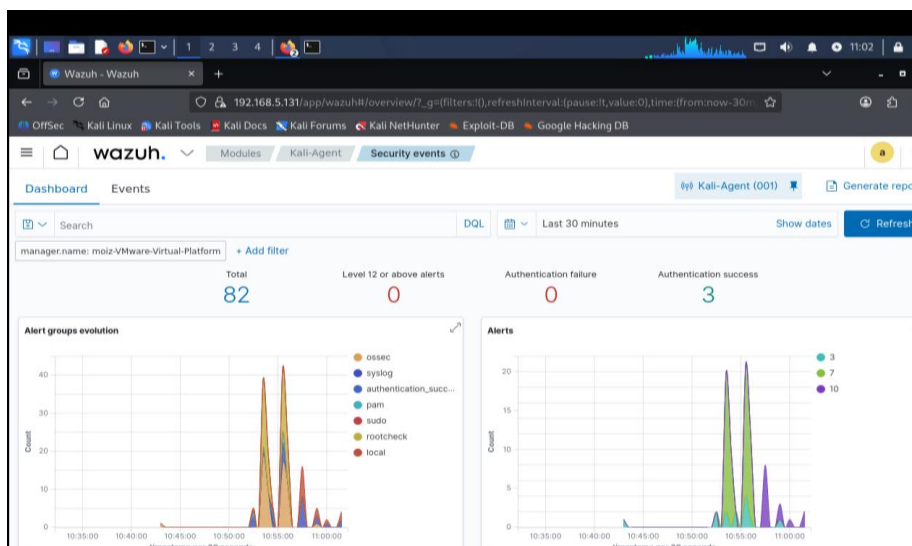
```
(moiz@Moiz)-[~]
$ sudo apt update && sudo apt install -y zip
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [20.9 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.9 MB]
56% [3 Contents-amd64 17.9 MB/52.9 MB 34%]
C

(moiz@Moiz)-[~]
$ whoami
moiz
```

Observation: The commands were executed without errors. The Wazuh Agent captured these terminal interactions through Auditd and streamed the telemetry to the Ubuntu Manager.

2. Real-Time Alert Monitoring (Machine: Wazuh Dashboard)

The Ubuntu Manager processed the logs and successfully matched them against our custom rules.



Observation: The "Alerts" graph shows significant peaks during the test window. The dashboard indicates multiple Level 10 alerts categorized under the 'syslog' and 'local' groups.

wazuh						Modules	Kali-Agent	Security events		
Security Alerts										
	Time ↓	Technique(s)	Tactic(s)	Description	Level	Rule ID				
>	May 15, 2026 @ 11:01:53.500			High Alert: 'whoami' command was executed on the agent!	10	100001				
>	May 15, 2026 @ 11:01:51.491			High Alert: 'whoami' command was executed on the agent!	10	100001				
>	May 15, 2026 @ 11:00:13.399			High Alert: 'whoami' command was executed on the agent!	10	100001				
>	May 15, 2026 @ 10:59:13.941			High Alert: Unauthorized package installation (apt install) detected!	10	100002				
>	May 15, 2026 @ 10:59:13.940			High Alert: Unauthorized package installation (apt install) detected!	10	100002				
>	May 15, 2026 @ 10:59:12.501			Log file rotated.	3	591				
>	May 15, 2026 @ 10:57:57.232			High Alert: Unauthorized package installation (apt install) detected!	10	100002				
>	May 15, 2026 @ 10:57:57.230			High Alert: Unauthorized package installation (apt install) detected!	10	100002				
.	May 15, 2026 @			High Alert: Unauthorized package installation (apt install)	..	100002				

Observation: The system successfully triggered both custom rules. As seen in the table, **Rule ID 100001** and **Rule ID 100002** are displayed with **Level 10 severity**, confirming that the Manager is correctly identifying both reconnaissance and unauthorized package installation attempts.